

Competitive Exam Practice 9: Cache Memory

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CS 1995 Q1.6, verified — GATEOverflow V3 p.35*] [**MCQ, 1 mark**] The principle of locality justifies the use of:

- (A) Interrupts
- (B) DMA
- (C) Polling
- (D) Cache Memory

Q2 [*GATE CS 2019 Q1, verified — GATEOverflow V3 p.45*] [**MCQ, 1 mark**] A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

- (A) 24 bits and 0 bits
- (B) 28 bits and 4 bits
- (C) 24 bits and 4 bits
- (D) 28 bits and 0 bits

Q3 [*GATE CS 2005 Q67, verified — GATEOverflow V3 p.37*] [**MCQ, 2 marks**] Consider a direct mapped cache of size 32 KB with block size 32 bytes. The CPU generates 32-bit addresses. The number of bits needed for cache indexing and the number of tag bits are respectively:

- (A) 10, 17
- (B) 10, 22
- (C) 15, 17
- (D) 5, 17

Q4 [*GATE CS 2014 Set-2 Q9, verified — GATEOverflow V3 p.43*] [**NAT, 2 marks**] A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____.

Q5 [*GATE CS 1995 Q2.25, verified — GATEOverflow V3 p.35*] [**MCQ, 2 marks**] A computer system has a 4 K word cache organized in block-set-associative manner with 4 blocks per set, 64 words per block. The number of bits in the SET and WORD fields of the main memory address format is:

- (A) 15, 40
- (B) 6, 4
- (C) 7, 2
- (D) 4, 6

Q6 [*GATE CS 2004 Q65, verified — GATEOverflow V3 p.37*] [**MCQ, 2 marks**] Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is:

8, 12, 0, 12, 8.

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q7 [*GATE CS 2015 Set-2 Q24, verified — GATEOverflow V3 p.43*] [**NAT, 1 mark**] Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

Q8 [*GATE CS 2014 Set-3 Q44, verified — GATEOverflow V3 p.43*] [**NAT, 2 marks**] The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____.

Q9 [*GATE CS 1992 Q5-a, verified — GATEOverflow V3 p.32*] [**NAT, 2 marks**] The access times of the main memory and the cache memory, in a computer system, are 500 ns and 50 ns respectively. It is estimated that 80% of the main memory requests are for read, the rest for write. The hit ratio for the read access only is 0.9 and a write-through policy (where both main and cache memories are updated simultaneously) is used. Determine the average time of the main memory (in ns).

Q10 [*GATE CS 2025 Set-1 Q43, verified — GATEOverflow V3 p.33*] [**NAT, 2 marks**] A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache; if not found it goes to L2 cache; if it fails there it goes to main memory, where the data is definitely available. The hit rates and access times are: L1 cache hit rate = 95%, access time = 10 ns; L2 cache hit rate = 85%, access time (including L1 cache miss penalty) = 20 ns; main memory access

time (including L1 and L2 cache miss penalties) = 200 ns. The average memory access time in nanoseconds (rounded off to two decimal places) is _____.

Q11 [*Original, GATE-pattern*] [**NAT, 2 marks**] Consider a fully associative cache using FIFO replacement. The reference string of block numbers is

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5.

The number of cache misses when the cache has 3 lines is _____, and when it has 4 lines is _____.

Q12 [*Original, GATE-pattern*] [**NAT, 2 marks**] Consider a fully associative cache with 3 lines, initially empty, using optimal (OPT / Belady's MIN) replacement. For the reference string

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2,

the number of cache *hits* is _____.

Q13 [*Original, GATE-pattern*] [**NAT, 2 marks**] Consider the same fully associative cache with 3 lines and the same reference string as Q12, but with LFU replacement (reference counts, ties broken by evicting the line resident longest). The number of cache *hits* is _____.